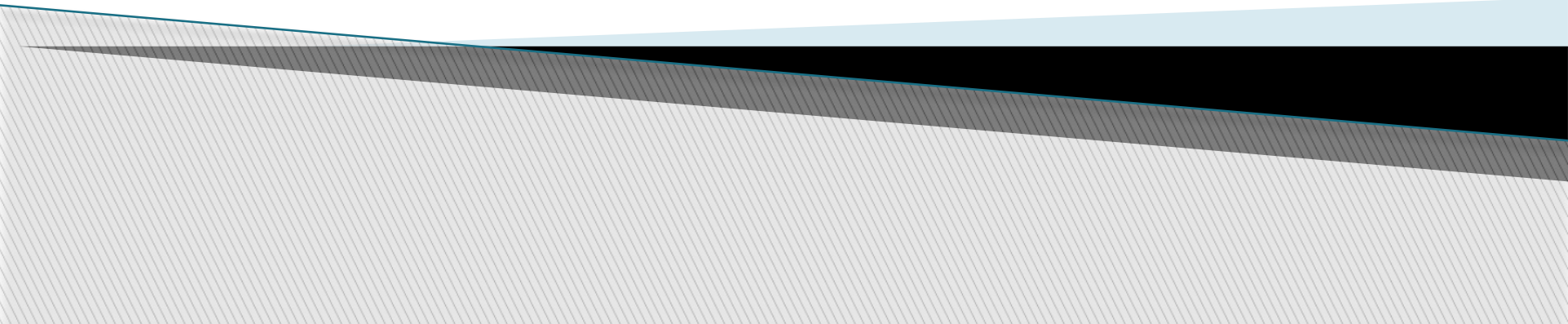
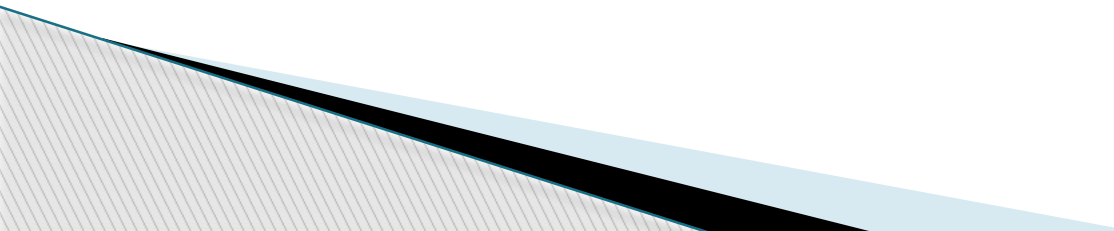


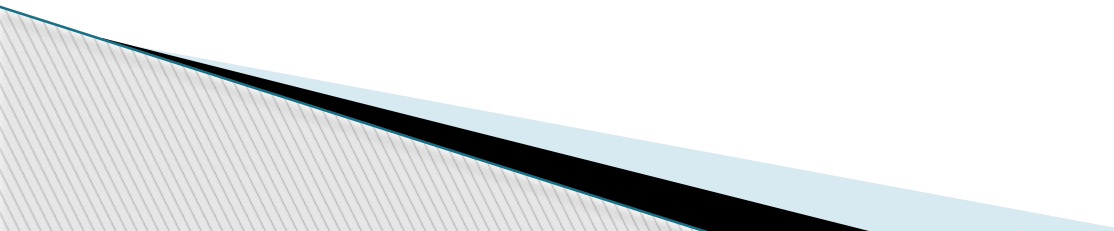
Internet Protocol (IP)



What is IP.....?

- ▶ IP stands for Internet Protocol
 - ▶ IP specifies the format of packets, also called datagrams, and the addressing scheme. Most networks combine IP with a higher-level protocol called Transmission Control Protocol (TCP), which establishes a virtual connection between a **destination** and a **source**.
- 

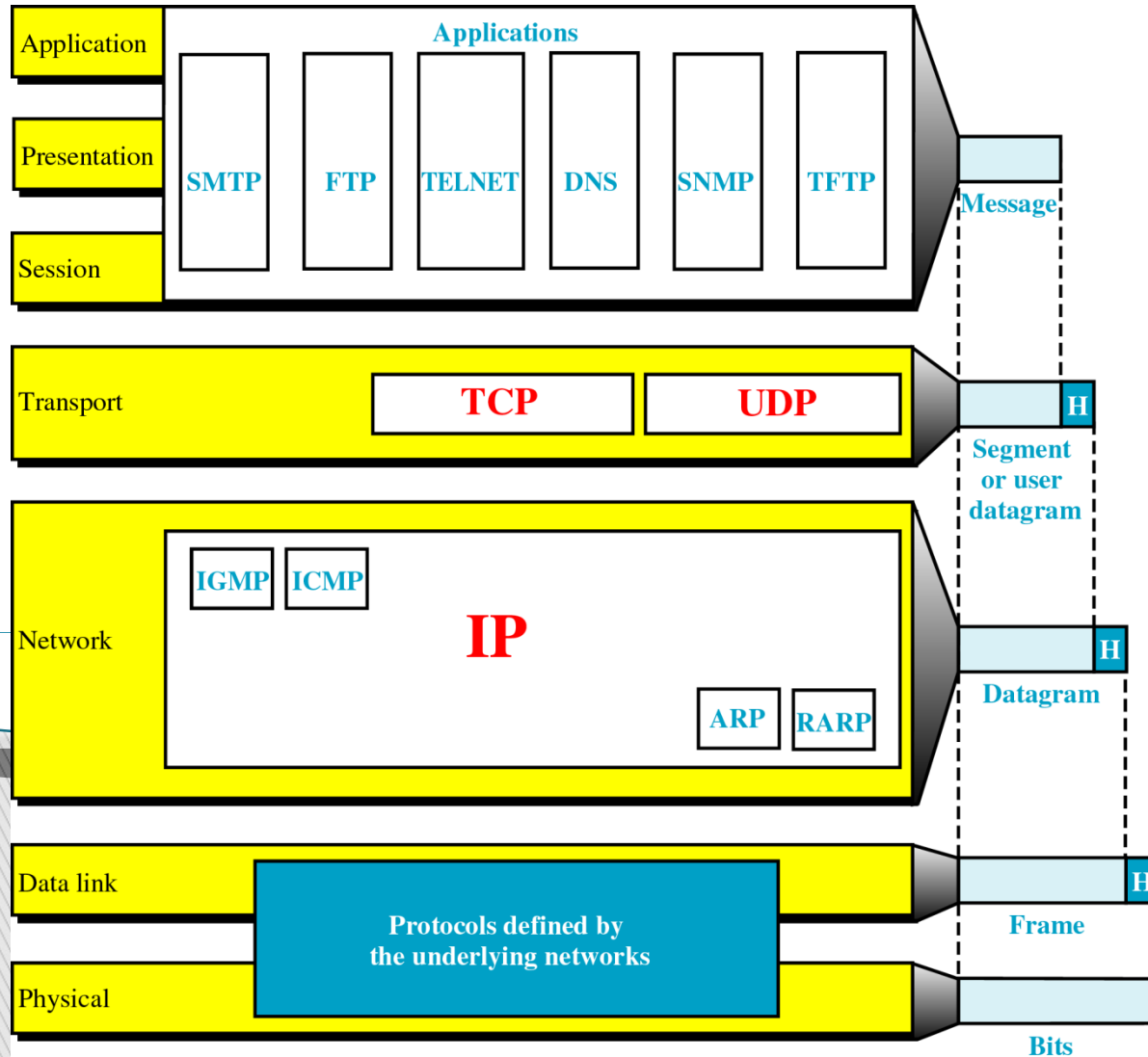
What is IP.....? (cont.)

- ▶ IP by itself is something like the postal system.
 - ▶ It allows you to address a package and drop it in the system, but there's no direct link between you and the recipient.
 - ▶ TCP/IP, on the other hand, establishes a connection between two hosts so that they can send messages back and forth for a period of time.
- 

Purpose of the IP....

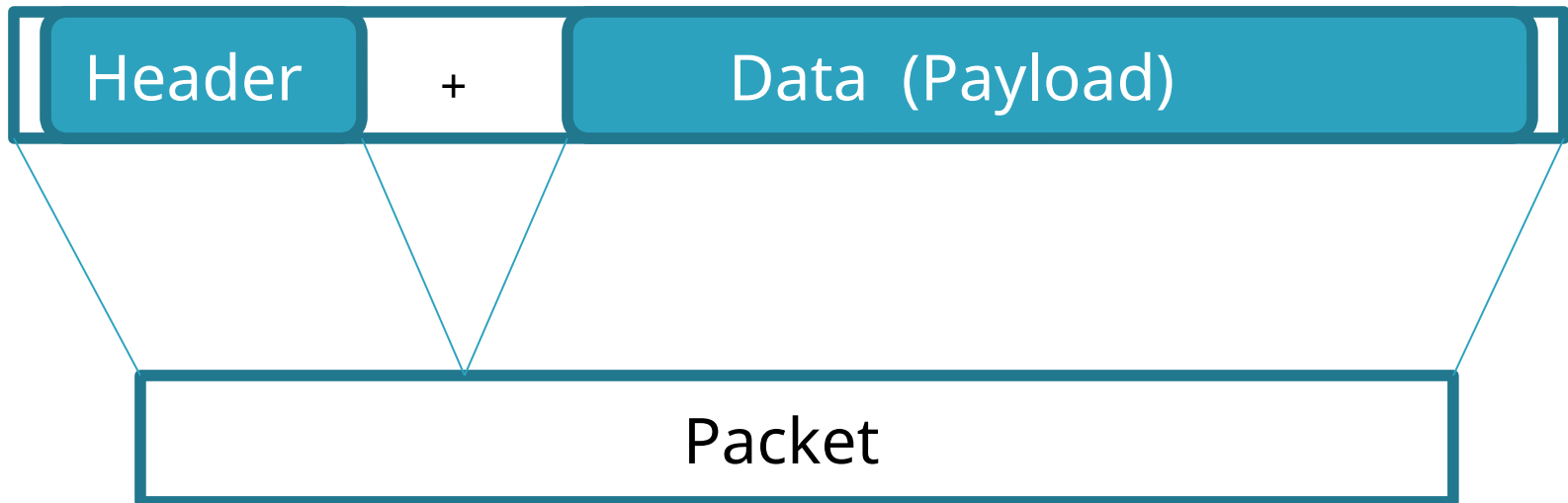
- ▶ The Internet Protocol defines the basic unit of data transfer (IP Datagram)
 - ▶ IP software performs the routing function
 - ▶ IP includes a set of rules that process the idea of unreliable packet delivery.
 - How hosts and routers should process packets
 - How & when error messages should be generated
 - The Conditions under which packets can be discarded.
- 

TCP/IP and the OSI Model



Construction of Datagrams....

- ▶ Each #datagram has two components
 - Header
 - Payload



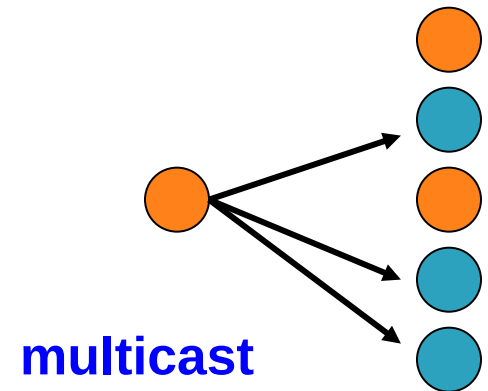
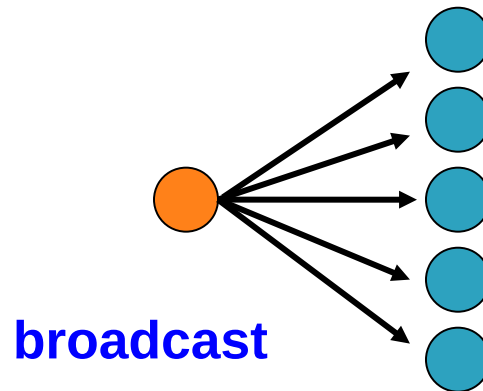
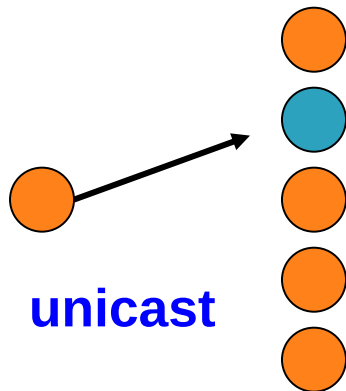
IP Service

- ▶ Delivery service of IP is minimal.
- ▶ IP provides an **unreliable connectionless** best effort service
 - **Unreliable** : IP doesn't make an attempt to recover lost packets
 - **Connectionless** : Each packet is handled independently
 - **Best Effort** : IP doesn't make guarantees on the service (No through output , No delay guarantee...)

IP Service (Cont....)

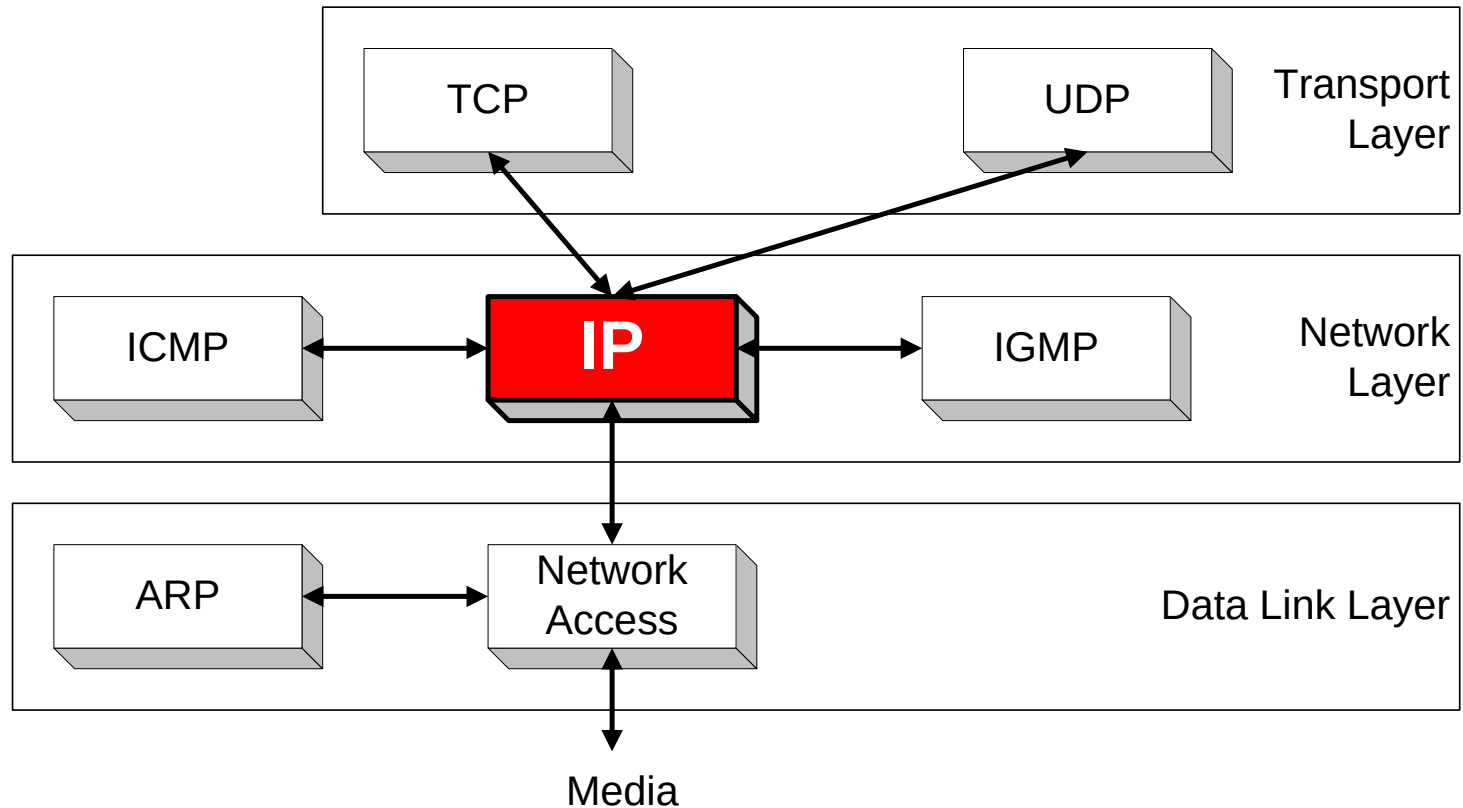
- ▶ IP supports the following services

- One-to-one (unicast)
- One-to-all (broadcast)
- One-to-several (multicast)



Orientation of Internet Protocol

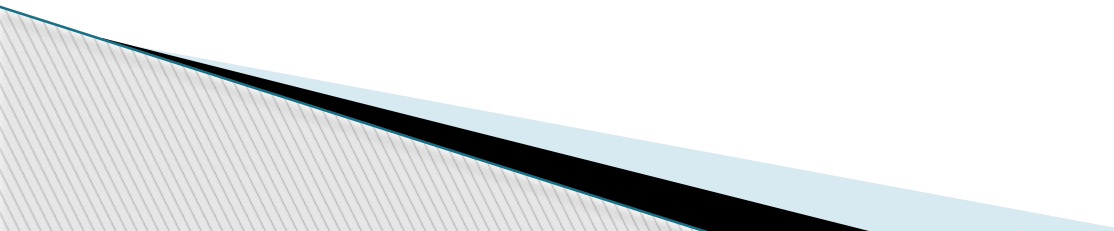
- IP is a **Network Layer** Protocol



Journey to IP Versions...

- ▶ IPV(1-3) : were not formally assigned.
- ▶ IPV4 : TCP/IP , 32bit IP address currently used.
- ▶ IPV5 : Internet Stream Protocol (SP)
 - ▢ Experimental Protocol
 - ▢ Never Introduced for public use.
- ▶ IPV6 : Designed to replace IPV4 , 128bit IP address

Features of IPV4...

- ▶ Connectionless protocol and best effort based.
 - ▶ Simplicity
 - ▢ It is simpler and easy to remember
 - ▢ Require less memory
 - ▶ Familiarity
 - ▢ Millions of devices are already knowing it
 - ▢ Existing infrastructure already support it
- 

IPv4 Header

VER 4 bits	HLEN 4 bits	Service 8 bits	Total length 16 bits	
Identification 16 bits			Flags 3 bits	Fragmentation offset 13 bits
Time to live 8 bits		Protocol 8 bits	Header checksum 16 bits	
Source IP address				
Destination IP address				
Option				

Version

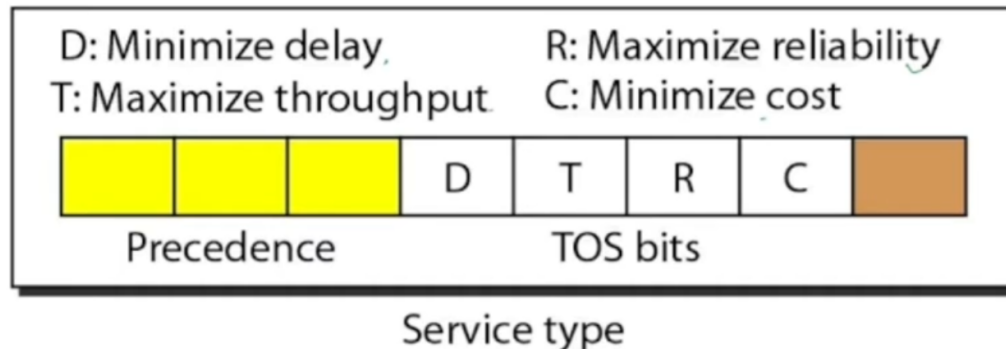
This Field defines the version of IP.

Header Length

- Header length is a 4 bit field that contains the length of the IP header.
- The length of IP header always lies in the range [20 bytes , 60 bytes]
- minimum length of IP header = $5 \times 4 \text{ bytes} = 20 \text{ bytes}$
- maximum length of IP header = $20 \text{ bytes} + 40 \text{ bytes} = 60 \text{ bytes}$.

Type Of Service

- Type of service is a 8 bit field that is used for Quality of Service (QoS).
- The datagram is marked for giving a certain treatment using this field.



Total Length

- Total length is a 16 bit field that contains the total length of the datagram (in bytes).

$$\text{Total length} = \text{Header length} + \text{Payload length}$$

- Minimum total length of datagram

20 bytes (20 bytes header + 0 bytes data)

- Maximum total length of datagram

65535 bytes

Identification

- Identification is a 16 bit field.
- It is used for the identification of the fragments of an original IP datagram.

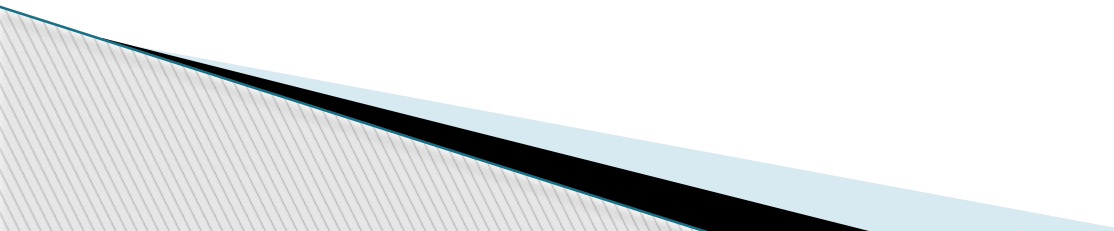
Flags:- 3 bits

- **First bit:** reserved, must be zero
- **Second bit:** (DF) 0 = May Fragment, 1 = Don't Fragment.
- **Third bit:** (MF) 0 = Last Fragment, 1 = More Fragments
- DF bit stands for Do Not Fragment bit.
- MF bit stands for More Fragments bit.

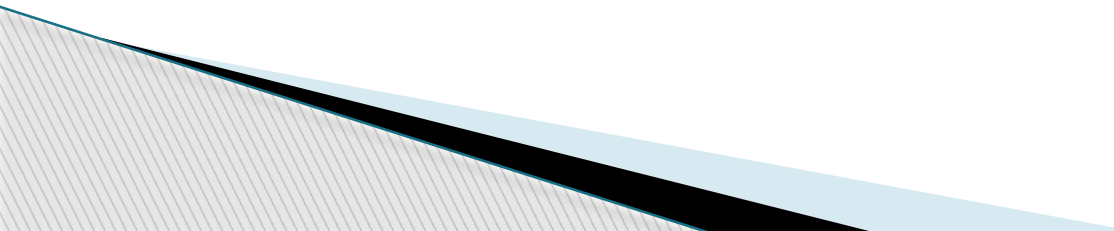
Fragment Offset

- Fragment Offset is a 13 bit field.
- Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram.

Time To Live-

- Time to live (TTL) is a 8 bit field.
 - It indicates the maximum number of hops a datagram can take to reach the destination.
 - The main purpose of TTL is to prevent the IP datagrams from looping around forever in a routing loop.
- 

Protocol

- Protocol is a 8 bit field.
 - It tells the next level protocol to the network layer at the destination side.
 - Protocol number of TCP is 6 and UDP is 17.
- 

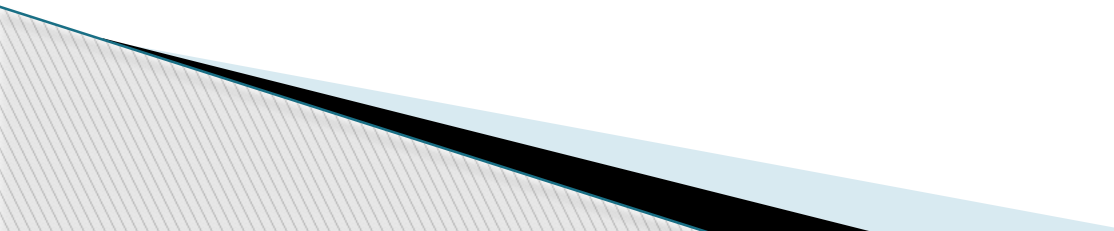
Header Checksum-

- Header checksum is a 16 bit field.
- It contains the checksum value of the entire header.
- The checksum value is used for error checking of the header.

Source IP Address

- Source IP Address is a 32 bit field.
- It contains the logical address of the sender of the datagram.

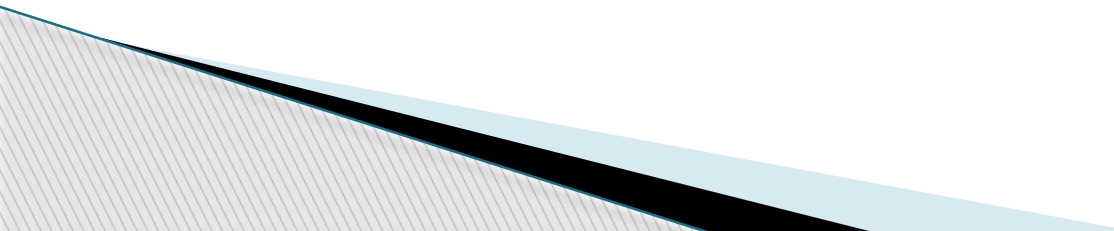
Destination IP Address

- Destination IP Address is a 32 bit field.
 - It contains the logical address of the receiver of the datagram.
- 

Options

- Options is a field whose size vary from 0 bytes to 40 bytes.
- This field is used for several purposes such as-
 1. Record route
 2. Source routing
 3. Padding

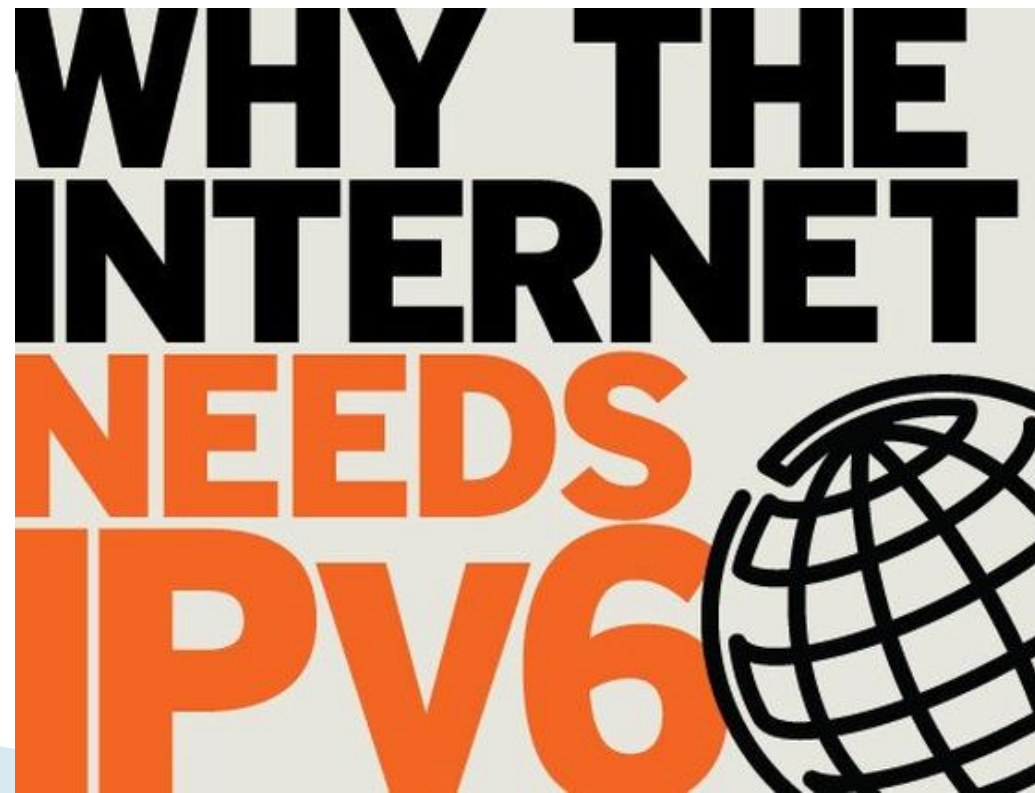
Benefits of IPV4....

- ▶ Widely support
 - ▶ Support of all Operating Systems
 - ▶ All commonly used protocols are supported
- 

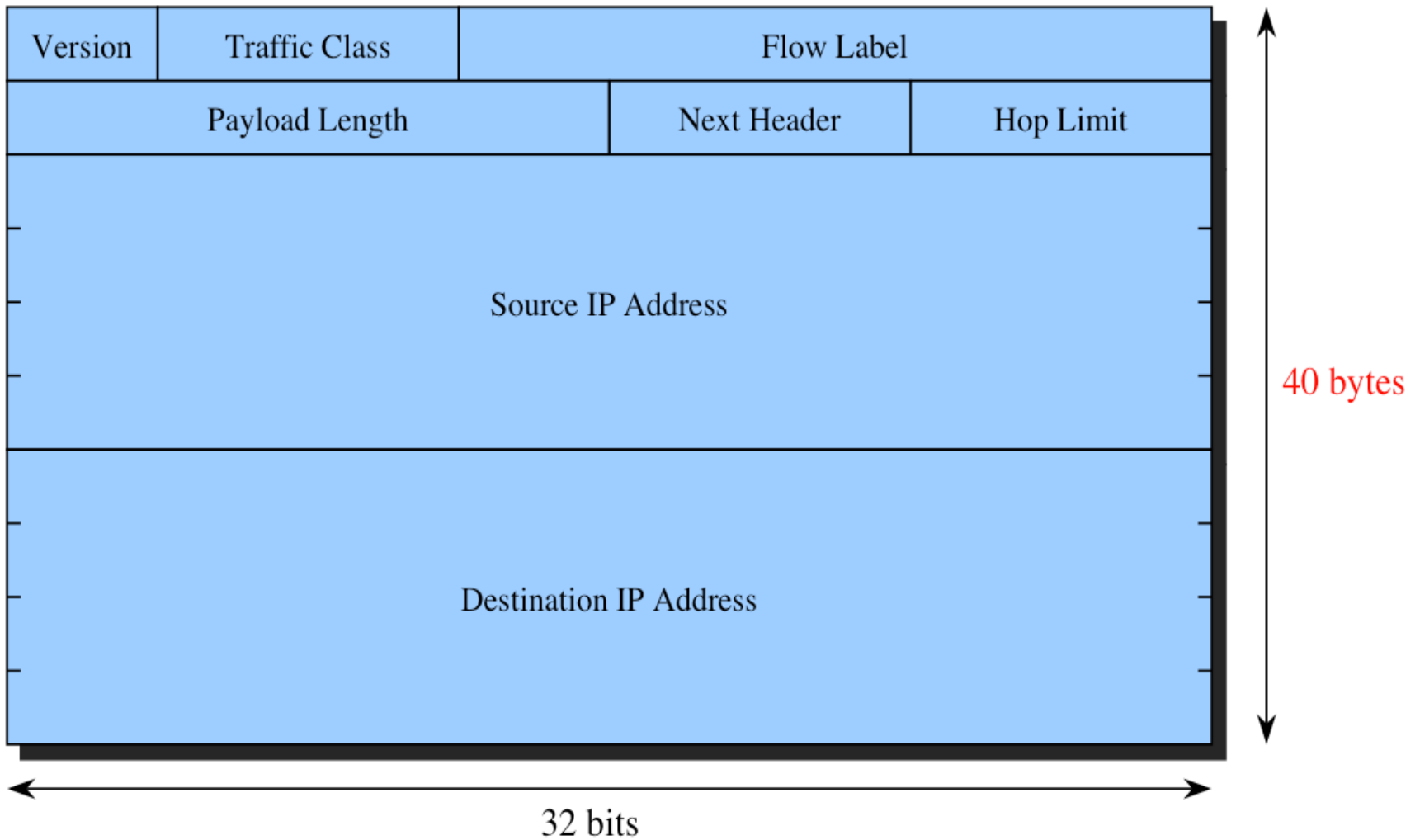
Why IPV6.....?

IPV6 provides a platform on new internet functionality that will be needed in the immediate future and provide

flexibility for future growth and **expansion**.



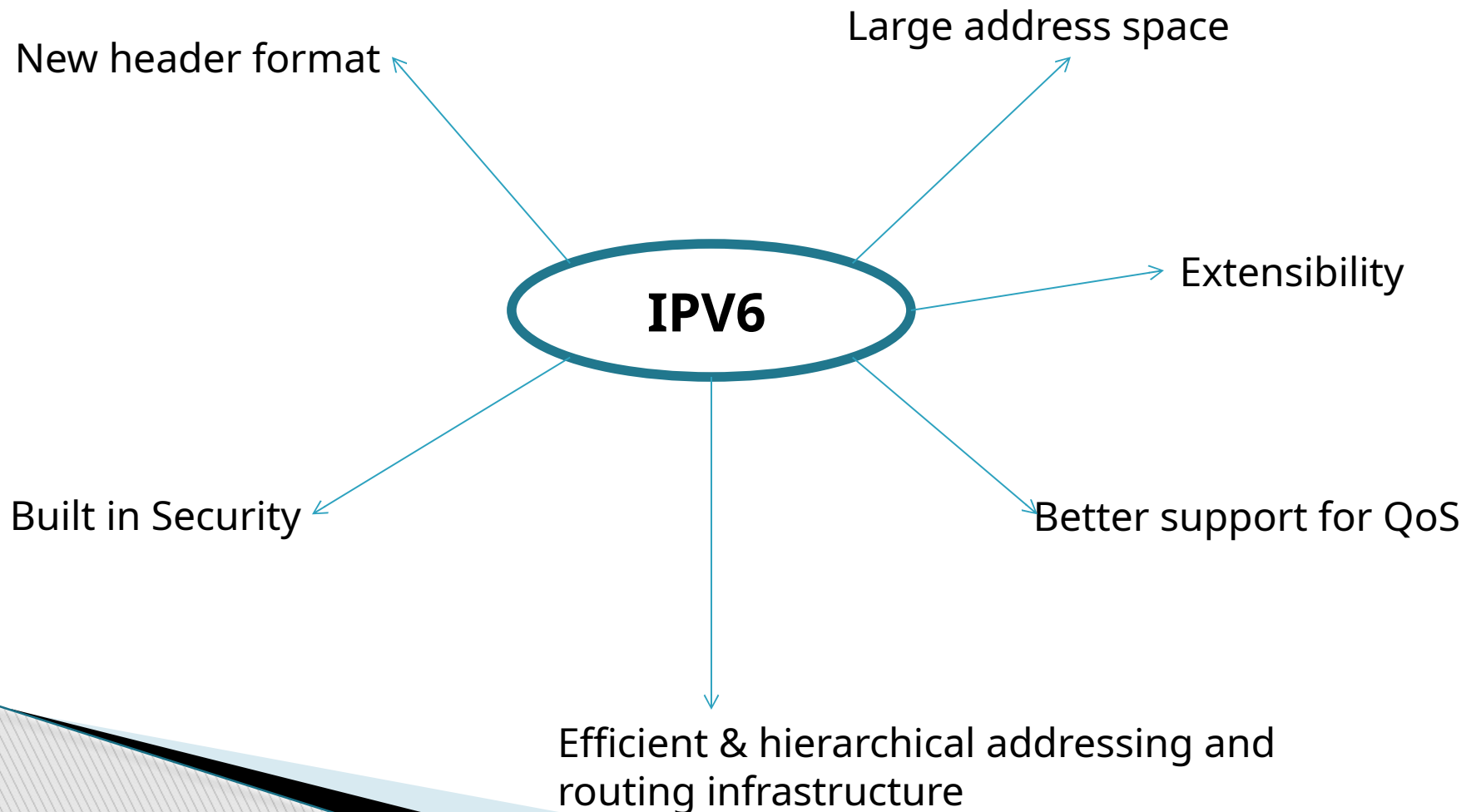
IPv6 Header



IPv6 Header Fields

- Version: IP version 6
- Traffic Class: used in congestion control similar to ToS field in IPv4.
- Flow Label: QoS management
- Payload length: payload length in bytes
- Next header: specifies the type of extension(if present)immediately following the IPv6 header
- Hop Limit: replaces the ttl field of IPv4
- Source and Destination addresses: 128 bits each

Benefits of IPV6.....



Difference between IPv4 & IPv6

<u>S.N</u>	<u>IPv4</u>	<u>IPv6</u>
1.	Addresses are 32 bits (4 bytes) long.	Addresses are 128 bits (16 bytes) long.
2.	Both routers & sending host fragment the packets.	Routers don't fragment the packets but sending host fragment the packets.
3.	Header includes a checksum.	Header doesn't include a checksum.

<u>S.N</u>	<u>IPv4</u>	<u>IPv6</u>
4.	Classes of addressing are A, B, C, D, E.	Classes of addressing are unicast, anycast, multicast.
5.	Configure either manually or through DHCP.	Doesn't require manual configuration.
6.	Must support a 576 byte packet size.	Must support 1208 byte packet size.

<u>S.N</u>	<u>IPv4</u>	<u>IPv6</u>
7.	IPv4 address uses the dot-decimal notation.	IPv6 address are represented in a hexadecimal, colon-separated notation.
8.	Not suitable for mobile networks.	IPv6 is better suited to mobile networks.
9.	Address space is small (2^{32}).	Larger address space (2^{128}).

<u>S.N</u>	<u>IPv4</u>	<u>IPv6</u>
10.	Internet protocol security(IPSec) is mandatory in this.	IPSec is optional.
11.	An IP address is made up of 4 bytes of information expressed as 4 number between 0 & 255 shown separated by periods. e.g.- 238.17.159.4	An IPv6 address is represented by 8 group of 16 bit hexadecimal values separated by colons (:). e.g.- 2001:0db8:85a3:0000:0000:8a2e:0370:7334

IP Based Technologies..

- ▶ Internet
 - ▶ VoIP
 - ▶ IP – TV
 - ▶ IP-VPN
 - ▶ Wireless Mobile Technology
 - ▶ Internet Broadcasting
- 